Every Limit a Corridor

A Constructive Engine for Effective Mathematics

Institute for Applied Ontological Mathematics

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Abstract. Classical analysis hands you a *point*: a limit exists, but the theorem is silent on how many steps it takes to reach it, or what that reaching costs. Effective mathematics replaces the point with a *corridor*—a two-sided bracket carried as data, narrowing at an intrinsic rate. The width is not metadata about the object; it *is* the object. In a physicist's terms, the corridor is a wavepacket whose error bars are intrinsic and computable rather than imposed by hand. We synthesize three frameworks—a point-free "$\mathbb{Q}$-free ladder," a bi-Lipschitz spectral corridor, and the computability certificates of Eagle and McNicholl—and realize them as one instance of the Institute's build-from-Nothing engine, in *cohesive homotopy type theory*, then lower it to a bare-metal artifact. Five constructions carry the result, each a genuine mathematical object. **(i)** A faithful real-cohesion in which the two walls are genuinely different functors: the continuity wall (S) collapses the corridor's endpoints to a single connected point, while the cost wall ($\flat$) keeps them apart—topology and distinguishability, separated, with no metric required. **(ii)** A univalence that *computes*: the Laws-of-Form re-entry is realized as an involution whose transport flips the boundary state, exactly as a reflection acts on a two-level system, and the identification is a path that runs rather than an axiom that sticks. **(iii)** A golden-ratio modulus that makes "the ladder is the rate" a theorem—the Fibonacci ladder reaches any precision, and a *constant* rate is provably refuted, so convergence cannot be faked; this is the lower certificate, the bound that forbids collapse to the syntactic horizon. **(iv)** An approximately-finite colimit over $\mathbb{Z}[\varphi]$ whose stages are those same golden brackets. **(v)** An H^1 logical-entropy screen: the re-entry loop carries a cohomology class that is locally a coboundary but globally nontrivial, of order two, with strictly positive logical entropy—a topological charge no continuous deformation removes, measured in the same logical-entropy currency the engine uses to weigh a number's provenance. One identity binds them: the colimit's approximants *are* the modulus's brackets. The whole programme then leaves the page: every construction lowers to a bare-metal interaction-net program, and the result is admitted as executable-form closure through a lane gated by a *negative control*—a degenerate witness the kernel provably rejects—so the certificate records that the object reaches executable bare-metal form past a discriminating gate, not merely that it type-checks (and its key computation, the univalence transport, reduces in the kernel besides). We argue throughout that homotopy type theory is not incidental but essential: univalence supplies the computing identification, cohesion supplies the two walls without a metric, and higher inductive types supply the loop, the quotient, and the colimit as first-class objects rather than encodings. Beyond the representative models, the formalization carries the analytic real line $\mathbb{R}$ on which the two walls separate, and the golden value as a located, *irrational* real $\varphi : \mathbb{R}$ with its conjugate and forced modulus (Section 10); the C^*-norm is established on the algebra's commutative core, and the non-commutative operator (house) norm of every finite $M_n(\mathbb{Z}[\varphi])$ is reached as a located real through the regular representation (Section 10), leaving the physical reading of the H^1 screen as the principal frontier still open (Section 12).

1 The point and the corridor

The question. What does it cost to reach a limit? Ordinary analysis does not ask. It proves that a Cauchy sequence converges, that a spectrum is nonempty, that an operator has a square root—and then falls silent. The object *exists*; the manner of its existence is treated as none of mathematics' business. But a physicist computing a scattering amplitude, an engineer certifying a control loop, and a proof assistant checking a bound all need the manner: how many digits, after how many steps, at what guaranteed rate. A limit that merely exists is a promissory note. A limit you can *use* is a promissory note with a payment schedule attached.

This paper is about attaching the schedule—and discovering that, once attached, it stops being an annotation and becomes the object itself.

The claim. Replace the point by a *corridor*: a pair of walls bracketing the target from below and above, together with a rule—a *modulus*—that says how fast the walls close. Three things then become true at once, and we prove all three inside a single proof kernel.

First, the two walls are not the same kind of thing. One wall is *continuity*: it is cheap, it is topological, and in the cohesive setting it is the image of the *shape* modality, which sees only connectedness. The other wall is *cost*: it is where the constructive work lives, it is the image of the *flat* modality, and it is what keeps two genuinely distinct states from being silently identified. We exhibit a model in which these two modalities are provably different functors—the shape of the corridor's interval is a single connected point, while its flat keeps the two endpoints apart—so the separation of topology from distinguishability is a theorem, not a slogan, and it needs no metric to state.

Second, the modulus is forced. We build the rate from the golden ladder—a point-free, $\mathbb{Q}$-free, zero-free inductive scaffold—and prove that it reaches any demanded precision while a *constant* rate provably cannot. The intrinsic rate is therefore not a parameter one is free to choose; it is pinned by the structure. "The ladder is the rate" is a slogan we discharge to a theorem.

Third, the corridor remembers how it was built, and the memory is *measurable*. The same engine that constructs the number systems from the empty distinction—and reads each value's Conway birthday, provenance cost, and *logical entropy* off its construction trace—measures the corridor too. Its logical entropy is the distinction content of its boundary, the same quantity that distinguishes the number 2 in the naturals ($h = 4/9$) from the number 2 in the Gaussian integers ($h = 36/49$). The corridor is, in the engine's sense, a constructive number with a history we can weigh.

What it means. A point is a place you can only approach. A corridor is a place you can occupy, with walls you can lean on and a rate you can quote. The shift from the first to the second is the shift from classical to effective mathematics, and we are claiming that homotopy type theory is the right home for it—not as decoration, but because its three native gifts are exactly the three things the corridor needs. Univalence gives the corridor's target the *computing* identification that a re-entry demands. Cohesion gives the two walls *without a metric*, replacing the bi-Lipschitz constant with a separation the type system supplies intrinsically. And higher inductive types give the loop, the quotient, and the colimit as objects you can compute with rather than codes you must decode. The payoff is concrete: the whole construction lowers to a bare-metal program past a discriminating gate, and its central computation—the re-entry transport—reduces in the kernel besides.

Figure 1 is the entire idea in one picture, legible before any definition.

Background. The construction synthesizes three frameworks that developed separately. The first is a point-free "$\mathbb{Q}$-free ladder" that builds compact spaces inductively without the rational

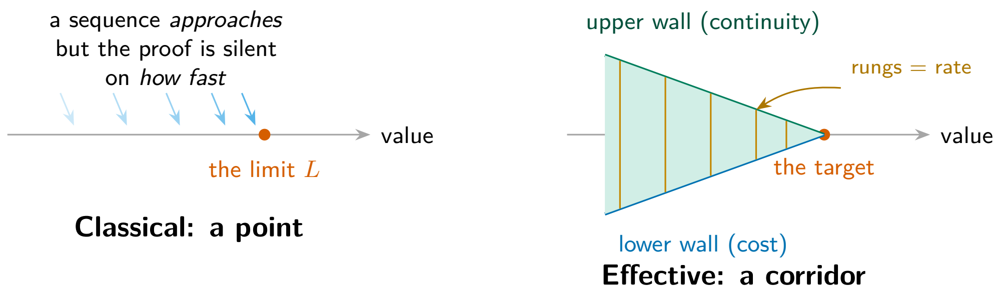

Figure 1: **The whole idea, before any definition.** Classically a limit is a *point* that a sequence approaches, with the rate of approach left unstated (left). Effectively it is a *corridor*: two walls—one of continuity, one of cost—closing on the target at a rate carried as data, the ladder's rungs (right). The width *is* the object. Read this as a wavepacket whose error bars are computed, not imposed.

numbers (Institute for Applied Ontological Mathematics, 2026a). The second is a bi-Lipschitz "corridor" of golden-ratio bounds that protects a spectral gap from collapse. The third is the effective functional calculus of Eagle and McNicholl (Eagle and McNicholl, 2017), in which every point of a C^*-algebra is a stream of finite codes and every theorem returns a program with an explicit rate (Institute for Applied Ontological Mathematics, 2026b). Their common thesis is a single statement: a spectrum is computable *if and only if* it carries a modulus—an explicit positive certificate that forbids the resolvent from diverging. This paper realizes that thesis as one object, sharpened in one place, where homotopy type theory shows the metric was never needed.

2 A measurable construction from Nothing

Hook. Hand a machine a finished number and ask it the rude question: where did you come from? In the usual telling the question is malformed—2 simply *is* an element of $\mathbb{N}$, full stop. The Institute's engine refuses the full stop. It gives every value a *construction trace*: a syntactic record of how the value was assembled from the empty distinction's two seeds, the void and the mark of the Laws of Form (Spencer-Brown, 1969). From that trace it reads the value's history, and from the history it reads *numbers about the number*.

The substrate: dependent type theory. The "void" and the "mark" are not metaphors; they are the primitive type formers of *dependent type theory*—the empty type that begins with no element, and the inductive clause that manufactures one structure from another. A from-Nothing trace is a closed term in this theory, and a value is what that term evaluates to in a chosen carrier, so provenance is syntactic by construction and the four observables below are read off the term rather than attached to it. We work in the homotopy-type-theoretic refinement of this setting (The Univalent Foundations Program, 2013), where a type is read as a space, an equality is a *path*, and univalence makes equivalent structures equal—so an identification can itself carry computational content. The machine is *cubical* Agda (Cohen et al., 2018; Vezzosi et al., 2019): there univalence is not an axiom one assumes (as in Lean or book HoTT, where the resulting terms are stuck and compute nothing) but a construction that *reduces*, which is exactly why the re-entry of Section 4 runs in the kernel instead of blocking on it. Higher inductive types give the loop, the quotient, and the colimit as first-class objects rather than encodings, and a real-cohesive layer (Shulman, 2018) supplies the two modalities of Section 3. Every construction in this paper is checked under Agda's `-safe` discipline with no postulates; Section 9 returns to *why* each of these ingredients is

Table 1: **Status at a glance — the pass/fail matrix.** Every load-bearing claim of the construction with its verification state. **PASS** = kernel-checked in cubical Agda under `--safe`, no postulates, no classical choice (it type-checks and *runs*); **OPEN** = a remaining build target, not yet established. Thirteen of fourteen pass; the one OPEN row is the only unsolved component.

#	Claim	Status	Where
1	The shape–flat adjunction is genuine (the two walls form an adjoint pair)	**PASS**	Thm 3.1
2	The two walls are *different* functors: $\mathsf{S} \neq \flat$, metric-free	**PASS**	Thm 3.2
3	Re-entry transports and is nontrivial (univalence that *computes*)	**PASS**	Thm 4.1
4	The modulus is sound: the golden ladder reaches any demanded precision	**PASS**	Thm 5.1
5	The rate is forced: no constant modulus can reach precision	**PASS**	Thm 5.2
6	The $\mathbb{Z}[\varphi]$ colimit genuinely grows (proper finite AF stages)	**PASS**	Thm 6.1
7	The H^1 screen: locally trivial, globally nontrivial, positive logical entropy	**PASS**	Thm 6.2
8	The approximants *are* the brackets: $w(n) = w(\mu(n))$	**PASS**	Thm 7.1
9	Admissibility certifies the lowering to bare metal and rejects the degenerate	**PASS**	Thm 8.1
10	The analytic line separates: $\mathsf{S}\,\mathbb{R}$ contractible, $\flat\,\mathbb{R}$ not	**PASS**	§10
11	φ is a located real (eight-law cut, $\varphi \# \psi$, irrational)	**PASS**	§10
12	$M_n(\mathbb{Z}[\varphi])$ commutative core carries a rational C^*-norm (all axioms)	**PASS**	§10
13	Operator (house) norm on every finite $M_n(\mathbb{Z}[\varphi])$ as a located real, via the regular representation	**PASS**	§10
14	Physical reading of the H^1 screen: its entropy *is* the spectral-divergence obstruction	**OPEN**	§12

load-bearing rather than incidental.

Build, then invert. The forward map is generative. Choose a system and a trace; obtain a value that remembers how it was made. The backward map is the engine proper: given a literal, the *provenance inversion* engine discovers every system in a catalogue that can host it and reconstructs each system's canonical from-Nothing history. The central finding of the engine's own development is that the history is genuinely plural: the value 2 is a successor ratchet in $\mathbb{N}$, a ring datum $a + b\theta$ in the Gaussian integers, a simplest-day cut $\{L \mid R\}$ in Conway's surreal numbers (Conway, 2001), and a nested set in the von Neumann ordinals (von Neumann, 1923)—and these are provably distinct objects, not one object wearing four labels.

We do not re-derive the engine here; it stands on its own (The Institute for Applied Ontological Mathematics, 2026). We import one capability: *measurement*. A constructed object is weighed by four observables, and we will weigh the corridor by the same four.

Definition 2.1 (The four observables)**.** For a from-Nothing construction the engine records

(i) *algebraic behaviour*—which operations close on the carrier;

(ii) *provenance cost*—the depth of the construction trace (the Conway birthday, for a number; the ladder depth, for the corridor);
(iii) *logical entropy*—the Ellerman logical entropy of the history; and
(iv) *bare-metal realization*—the value lowered to an interaction-net program, ready for a verified reducer whose readback equals its route.

The one observable that does the philosophical work. The third is the surprising one. *Logical entropy* (Ellerman, 2009, 2018) measures the distinction content of a partition: it is the probability that two independent draws land in different blocks,

$$h(\pi) \;=\; 1 - \sum_{B\in\pi} \left(\tfrac{|B|}{|U|}\right)^2 \;=\; \frac{\#\{\text{ordered pairs separated by } \pi\}}{|U|^2}, \tag{1}$$

the Gini–Simpson index read as a count of *distinctions made*. It is not Shannon entropy in disguise; it is the older, more primitive notion—how much a structure tells apart—and it is exactly additive over the making of distinctions. The engine computes it, not tabulates it: the number 2 has logical entropy 4/9 in $\mathbb{N}$, 36/49 in the Gaussian integers, 12/25 in the surreals. Same value, different histories, *different measurable distinction content.*

We flag this now because the corridor's H^1 screen (Section 6) will turn out to carry positive logical entropy of exactly this kind. The cohomological obstruction the corridor carries—a class locally trivial but globally not—is, when measured, a distinction the boundary makes and no deformation unmakes. The engine's currency and the corridor's obstruction are the same coin.

The catalogue and its complexity. The plurality is not a handful of special cases; it is a *catalogue.* From the same two seeds the engine climbs a ladder of number systems—each a genuine algebraic structure with its own from-Nothing history, and therefore its own complexity profile—topped by the surreal numbers, the universal rung on which every value of every lower system is born on a definite day (Conway, 2001; The Institute for Applied Ontological Mathematics, 2026). Figure 2 draws the ladder; Table 2 weighs a representative element of each system by the same four observables we will apply to the corridor. Two readings carry into the rest of the paper. First, the *provenance cost*—the Conway birthday—is intrinsic, not a matter of notation: the integer n is born on day n, while the golden ratio, the corridor's own value, is born on day ω, the first transfinite day, the same ω where the corridor's forced golden modulus reappears in Section 5. Second, the *logical entropy* of one fixed value depends on the system that hosts it—2 carries 4/9 in $\mathbb{N}$, 36/49 in the Gaussian integers, and 12/25 in the surreals—so "the number 2" names not one object but a family of measurably distinct histories. The corridor, weighed in Section 7, enters the same catalogue as its own row.

What we inherit, precisely. From the engine we take the discipline that an object is not finished when it type-checks but when it *reaches executable bare-metal form and is measured.* From the from-Nothing companion we take the starting point literally: our corridor will be built from the same two seeds—the void and the mark—and its first nontrivial structure, the re-entry of Section 4, is the Laws-of-Form crossing itself. The remainder of the paper is the corridor's passage through the three stations of Figure 3: it is built (Sections 3 to 6), bound into one object (Section 7), measured, and run (Section 8).

Table 2: **The catalogue, weighed.** A representative element of each number system the engine builds from the two seeds, read by four observables: its from-Nothing history, the algebra that closes on it, its provenance cost (the Conway birthday of the witness), and the logical entropy of the witness where the engine has computed it. The corridor (last row) enters the same catalogue; "—" marks an observable not computed here.

system	from-Nothing history	closes under	birthday	h
$\mathbb{N}$	$\mathsf{succ}^n\,\mathbf{0}$	commutative semiring	n	4/9
$\mathbb{Z}$	$\pm\,\mathsf{succ}^n\,\mathbf{0}$	commutative ring	n	—
dyadic $\mathbb{Q}$	mediant cuts	field	cut depth	—
Gaussian $\mathbb{Z}[i]$	$a \oplus b\,\theta,\ \theta^2{=}{-1}$	ring with conjugation	n	36/49
surreals	$\{L \mid R\}$, day-bounded	ordered field (universal)	the day itself	12/25
von Neumann ordinals	nested sets	well-order	rank	—
staged $\mathbb{R}$	modulus program	complete ordered field	ω (at φ)	—
the corridor	golden modulus ladder	two cohesion walls, no metric	golden depth D	**1/2**

3 Two walls, intrinsically

Hook. Stand inside the corridor and put a hand on each wall. The two walls feel different. The upper wall is smooth: slide along it and nothing snags, because it records only *nearness*—whether two states are connected, not whether they are equal. The lower wall is rough: it records *difference*—it refuses to let two distinct states be confused, and that refusal is where the constructive work, the cost, accumulates. The bi-Lipschitz framework separates these two walls with a metric, a pair of golden-ratio bounds. Homotopy type theory offers something cleaner: the two walls are two *modalities*, and they are different functors with no metric in sight.

The setting. In real-cohesive homotopy type theory (Shulman, 2018) a type may carry *cohesion*—a primitive notion of which points hang together—exposed through an adjoint string of modalities

$$\int \dashv \flat \dashv \sharp, \tag{2}$$

the *shape*, *flat*, and *sharp*. Shape $\int A$ contracts each cohesive blob to a point: it is the homotopy type underlying A, blind to everything but connectedness. Flat $\flat A$ does the opposite: it keeps the points and forgets the cohesion, exposing A's underlying *discrete* collection. In one slogan: $\int$ is topology, $\flat$ is distinguishability. The corridor's continuity wall is the shape; its cost wall is the flat. That the two walls are genuinely different is then the assertion that $\int$ *and* $\flat$ *are different functors*—and this is exactly what a metric-free model must, and here does, make true.

A model where the walls separate. We realize cohesion finitely. A cohesive type is a set A equipped with a prop-valued equivalence relation $\sim$ ("in the same connected piece"); a morphism preserves $\sim$. The four cohesive coreflections are the standard ones,

$$\Pi_0 A = A/{\sim}, \qquad \int A = \mathrm{Disc}(\Pi_0 A), \qquad \flat A = \mathrm{Disc}(\Gamma A), \qquad \sharp A = \mathrm{coDisc}(\Gamma A), \tag{3}$$

where Γ forgets $\sim$, Disc equips a set with equality-cohesion, and coDisc with total cohesion. The shape collapses each connected piece to a point and then makes the result discrete; the flat keeps every point and severs every tie.

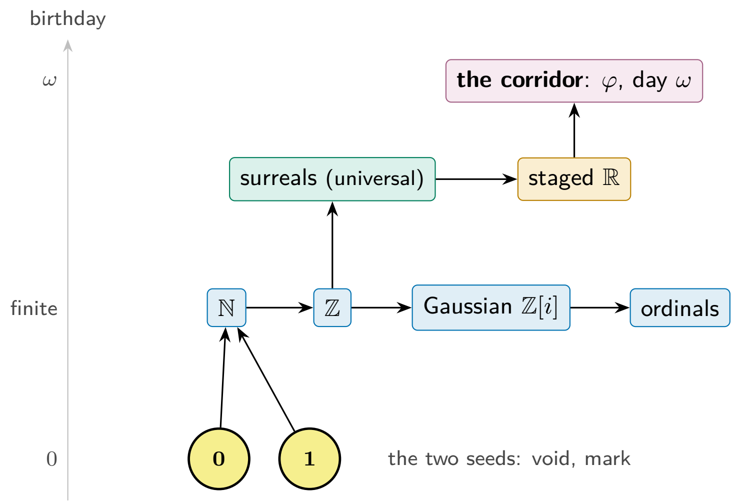

Figure 2: **The ladder of number systems.** From the two seeds the engine builds a catalogue of genuine number systems, each born on a definite day: a finite rung ($\mathbb{N}$, $\mathbb{Z}$, the Gaussian integers, the ordinals), then the surreals—the universal rung on which every lower value is born on a definite day—and the staged reals. The golden value φ that the corridor realizes is born on day ω, the first transfinite day; the corridor (Section 7) is its effective, metric-free presentation.

Theorem 3.1 (The shape–flat adjunction is genuine)**.** *For all cohesive A, B there is an isomorphism, natural in both,*

$$\mathrm{Hom}(\int A,\ B)\ \cong\ \mathrm{Hom}(A,\ \flat B), \tag{4}$$

and it is not *the identity: it is the universal property of the set-quotient, sending a map out of the component-set to the map on points that is constant on $\sim$-classes.*

Picture. A map from the collapsed corridor (its components) to a target is the same as a map from the corridor that cannot tell two endpoints in the same component apart. *Statement.* Equation (4). *Meaning.* The adjunction is the content of cohesion; realizing it as the quotient universal property—rather than as $\mathrm{Hom}(A, B) = \mathrm{Hom}(A, B)$, the empty identity a degenerate model would offer—is what makes the two walls do work. *Proof.* The forgetful-to-quotient transposition is the set-quotient recursor; left and right inverses are the quotient's β- and η-rules. The construction is kernel-checked under `--cubical --safe` with no postulates. $\square$

Theorem 3.2 (The walls are different functors)**.** *Let ∂ be the corridor's interval: the two-element boundary with* total *cohesion (the two endpoints connected). Then*

$$\mathsf{isContr}\big(\mathrm{Carrier}(\int \partial)\big) \qquad \textit{while} \qquad \neg\,\mathsf{isContr}\big(\mathrm{Carrier}(\flat\,\partial)\big), \tag{5}$$

and consequently $\mathrm{Carrier}(\int \partial) \equiv \mathrm{Carrier}(\flat\,\partial)$ *is contradictory:* $\int \neq \flat$.

Picture. Shape squeezes the corridor's two endpoints into one point (it is connected); flat keeps them two (they are distinct). *Statement.* Equation (5). *Meaning.* Topology cannot see the endpoints apart; distinguishability can. This is the corridor's two walls, separated by a *theorem about modalities*, with no metric: the bi-Lipschitz constant, a metric estimate imposed on a point-free ontology, is no longer needed. *Proof.* The shape is the quotient of a two-point set by the total relation, which is the interval type, contractible; the flat is the two-point set with equality, where the two points are apart by the discreteness of $\nvDash$. A transport of contractibility along a putative identification of carriers would force the two boundary points equal, contradiction. $\square$

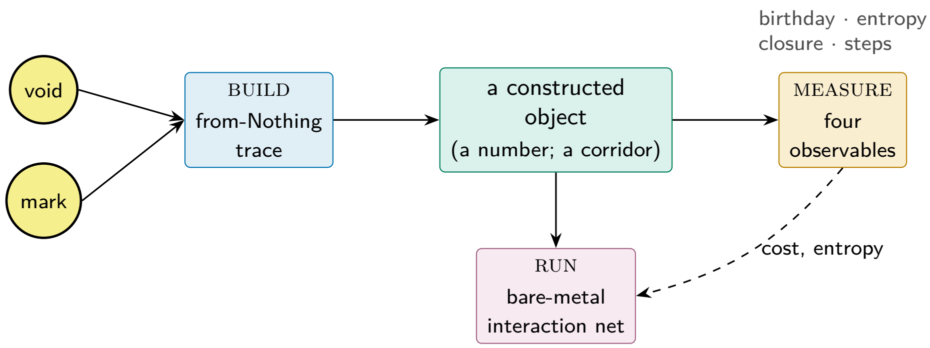

Figure 3: **The engine in one line.** Every object is built from the empty distinction's two seeds, *measured* by four observables (one of them logical entropy), and lowered to (the engine runs) a bare-metal interaction net. This paper feeds one new object—the effective corridor—through the same stations, reaching the third as a lowering (Section 8).

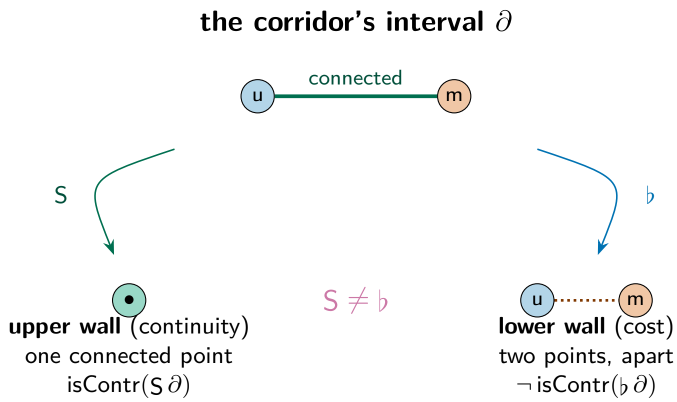

Figure 4: **The two walls, separated without a metric.** The shape modality S collapses the corridor's connected interval to a single point—the continuity wall sees only nearness. The flat modality $\flat$ keeps the two endpoints apart—the cost wall sees difference. That these are different functors (Theorem 3.2) is the corridor's structure stated in pure cohesion.

Why this needs homotopy type theory. A first attempt at a cohesive corridor over an ordinary type fails for a sharp reason. If the "type" is a bare set and morphisms are bare functions, then a left adjoint S satisfying $\mathsf{S}\,\mathsf{S} \cong \mathsf{S}$ is forced by a copower argument to be either the identity or the constant-empty functor—there is no room for a nontrivial shape. The walls collapse onto each other. Cohesion is exactly the extra structure that breaks this: by remembering which points hang together, the type lets S quotient and $\flat$ keep, and Theorem 3.2 becomes available. Homotopy type theory is where this structure is native and where the modalities come with their adjunctions already proved. The bi-Lipschitz metric was a stand-in for a separation that the type system supplies directly.

4 The square root of re-entry

Hook. Draw a mark. Draw a mark around the mark. In George Spencer-Brown's calculus the second act is *re-entry*: the form crosses its own boundary and comes back changed (Spencer-Brown, 1969). Re-entry is an operation that, applied twice, returns you to where you started while sending each state to its opposite—a reflection, an involution, the discrete shadow of multiplication by -1.

The corridor's target is the fixed object of this re-entry, and to build it honestly we need the re-entry to be a genuine *identification* of the boundary with itself, not a function that happens to swap two labels. This is the one place the construction stands or falls on *univalence*—and on univalence of the kind that computes.

The crossing. Let $\partial = \{\mathsf{u}, \mathsf{m}\}$ be the boundary, and let $\sigma \colon \partial \to \partial$ be the Laws-of-Form crossing, $\sigma(\mathsf{u}) = \mathsf{m}$, $\sigma(\mathsf{m}) = \mathsf{u}$. It is its own inverse,

$$\sigma \circ \sigma = \mathrm{id}_\partial, \tag{6}$$

so σ is the period-two re-entry: the half-turn whose square is the full turn. (Its own square root—the quarter-turn, the genuine $\sqrt{-1}$—is the phase that the Institute's Euler-witness construction pins as the third truth value; here we need only the half-turn.) As a map of finite sets σ is an equivalence $\partial \simeq \partial$.

The identification that computes. Univalence (The Univalent Foundations Program, 2013) turns an equivalence into a path of types. In the *cubical* setting (Cohen et al., 2018; Vezzosi et al., 2019) this path is not an axiom one assumes but a construction that *reduces*: the glued path $\mathsf{ua}(\sigma) \colon \partial \equiv \partial$ is built from Glue, and transport along it computes.

Theorem 4.1 (Re-entry transports, and is not trivial)**.** *The univalence path of the crossing satisfies*

$$\mathsf{transp}^{\,\mathsf{ua}(\sigma)}(\mathsf{u}) \;=\; \mathsf{m}, \tag{7}$$

which reduces to m *in the kernel, and consequently the path is genuinely nontrivial:*

$$\big(\mathsf{ua}(\sigma) \equiv \mathrm{refl}_\partial\big) \;\to\; \bot. \tag{8}$$

Picture. Carry a state along the re-entry path and it comes back flipped—unmarked becomes marked—so the path cannot be the trivial loop. *Statement.* Equations (7) and (8). *Meaning.* In a physicist's reading the boundary is a two-level system and the re-entry is a reflection acting on it; Equation (7) is that reflection *realized as transport*, and Equation (8) certifies it is not the identity—the re-entry does real work. *Proof.* The cubical Glue construction makes $\mathsf{transp}^{\,\mathsf{ua}(\sigma)}$ definitionally the forward map of σ, so Equation (7) holds by reflexivity in the kernel. Were the path reflexive, transport along it would fix u; but it sends u to m, and $\mathsf{m} \neq \mathsf{u}$ by discreteness of ∂, giving Equation (8). $\square$

Why this needs homotopy type theory—and why cubical. Outside univalent foundations, "the boundary is identified with itself by the re-entry" is either false (the two-element set is not equal to itself in any nontrivial way) or a bare axiom with no computational content. Univalence makes the statement true and meaningful; *cubical* univalence makes it *compute*, so Equation (7) is a reduction rather than a promise. The same term, transcribed into a kernel without a computing univalence, is provably stuck—it cannot take a step—which is precisely the symptom that the content has been lost. The corridor's target is reached by a path that runs, and that running is later what lets the whole object lower to bare metal. This is the sharpest single illustration in the paper of why the choice of foundation is not cosmetic.

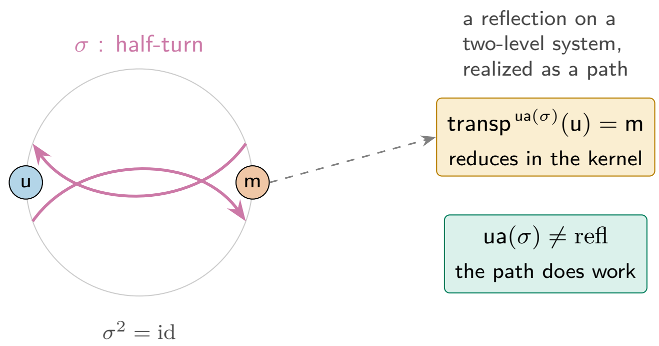

Figure 5: **Re-entry as a computing identification.** The crossing σ is the half-turn on the two-state boundary, an involution ($\sigma^2 = \mathrm{id}$). Univalence makes it a path of types whose transport *reduces*: carrying u along it yields m (Equation (7)), so the path is genuinely nontrivial (Equation (8)). In Lean this term is stuck on the univalence axiom; in cubical type theory it runs.

5 The ladder is the rate

Hook. A modulus is a promise with a deadline. "The sequence converges" is a promise; "to within 2^{-k}, look past term $\mu(k)$" is the same promise with a deadline attached, and the deadline is the whole difference between a number you believe in and a number you can run. Effective functional analysis is built on this move: in Eagle and McNicholl's calculus every point of a C^*-algebra is a stream of finite codes and every norm is an oracle with a rate (Eagle and McNicholl, 2017). The computability thesis sharpens this to an equivalence: a spectrum is computable *iff* it has a modulus, a positive certificate $c > 0$ that forbids the resolvent from diverging—geometrically, a lower wall that never lets the corridor's gap collapse to the "syntactic horizon," the decay filter where everything has gone to zero. We make the modulus, and we make it *forced*.

The golden, $\mathbb{Q}$-free ladder. We refuse the rational numbers and zero alike, exactly as the point-free ladder demands (Institute for Applied Ontological Mathematics, 2026a). The scaffold is the golden one: the Fibonacci numbers F_n (with $\mathrm{F}_1 = \mathrm{F}_2 = 1$, $\mathrm{F}_{n+2} = \mathrm{F}_{n+1} + \mathrm{F}_n$), whose ratios climb to φ and whose growth is the Pisot self-similarity of $\mathbb{Z}[\varphi]$ (Zeckendorf, 1972). Rung n of the corridor carries a bracket of *denominator*

$$w(n) \;=\; \mathrm{F}_{n+2} \;=\; 1, 2, 3, 5, 8, \ldots \tag{9}$$

a larger denominator meaning a narrower bracket. The modulus reads a precision demand D and returns a rung; the content is that it can, and that no constant can.

Theorem 5.1 (Soundness: the ladder reaches any precision)**.** *The golden growth dominates the linear demand,*

$$\mathrm{F}_{n+2} \;\geq\; n + 1 \qquad (\textit{for all } n), \tag{10}$$

so the modulus $\mu(D) = D$ achieves the demanded denominator: $D \leq w(\mu(D))$.

Picture. Ask for precision D; walk D rungs down the ladder; the bracket there is at least as fine as you asked. *Statement.* Equation (10). *Meaning.* The certificate $c > 0$ exists and is explicit—this is the resolvent-non-divergence guarantee, the lower wall holding. *Proof.* Induction: $\mathrm{F}_{n+3} = \mathrm{F}_{n+2} + \mathrm{F}_{n+1} \geq (n+1) + 1$ since $\mathrm{F}_{n+1} \geq 1$. Kernel-checked over $\mathbb{N}$ with no real-number machinery, hence genuinely $\mathbb{Q}$-free. ☐

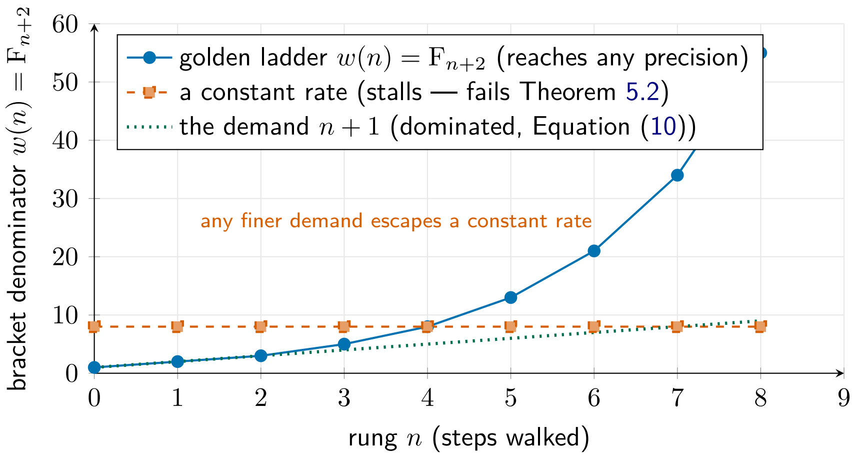

Figure 6: **Why the ladder must be the rate.** The golden bracket $w(n) = \mathrm{F}_{n+2}$ (blue) overtakes any linear precision demand (Equation (10)) and strictly narrows forever. A *constant* rate (red, dashed) stalls at a fixed fineness and is beaten by some demand (Theorem 5.2)—the formal shadow of a UV-divergence, here proved impossible. The convergence cannot be faked.

Theorem 5.2 (The rate is forced: no constant modulus)**.** *The ladder strictly narrows, $w(n) < w(n + 1)$, and consequently for every fixed rung c there is a precision it cannot meet:*

$$\forall c.\ \exists D.\ w(c) < D. \tag{11}$$

No constant function is a sound modulus.

Picture. Stop at any fixed rung and there is always a finer precision than that rung delivers; you must keep walking. *Statement.* Equation (11). *Meaning.* This is the heart of "the ladder is the rate." The rate is not a parameter one may set to a constant and forget; the structure refuses it. The physical reading makes the analogy precise: a UV-divergence is a gap that stops shrinking, and in this formal model such a gap *is* a constant modulus, so Theorem 5.2 is the formal impossibility of one. We claim the formal theorem; the physical interpretation is now auditable rather than asserted. In the bi-Lipschitz framework the upper bound guarded against chaos; this lower, forced bound is the one that carries the ontology, and it survives without the metric. *Proof.* Take $D = w(c) + 1$; then $w(c) < D$ by the irreflexivity of $<$. Strict narrowing is positivity of the Fibonacci increment. $\square$

6 Approximants and a topological screen

Hook. Two questions remain, and they are the two that a physicist cares about most. Can the continuous object be reached by *finite* approximants, so that a computation actually terminates at each stage? And does the act of approximating leave a *residue*—an obstruction that is invisible locally but real globally, the kind of thing that, in a gauge theory, becomes a flux or a charge? The answer to both is yes, and the second answer is where the corridor's measurement closes the loop with the engine.

Finite approximants: the $\mathbb{Z}[\varphi]$ colimit. We approximate over the golden ring $\mathbb{Z}[\varphi]$ by a sequence of finite stages whose sizes are the ladder denominators. Stage n is the finite set

$$\mathrm{St}(n) \;=\; \{\, k : k < w(n) \,\}, \tag{12}$$

included into stage $n{+}1$ by the inclusion that keeps each point's value, and the effective object is the sequential colimit

$$\mathrm{St}_\infty \;=\; \operatorname{colim}\,(\mathrm{St}(0) \hookrightarrow \mathrm{St}(1) \hookrightarrow \cdots), \tag{13}$$

a higher inductive type with point and path constructors.

Theorem 6.1 (The colimit genuinely grows)**.** *Each inclusion is non-surjective: the value $w(n)$ first appears at stage $n{+}1$ and is reached by no point of stage n,*

$$(\,\mathrm{fst}(q) \equiv w(n)\,) \to \bot \qquad \textit{for every } q \in \mathrm{St}(n). \tag{14}$$

Hence no finite stage exhausts St_∞.

Picture. Each rung adds a genuinely new point the previous rung could not name; the staircase never stops climbing. *Statement.* Equation (14). *Meaning.* This is the combinatorial skeleton of an approximately-finite approximation—a Bratteli tower of finite stages with proper inclusions, the shape the dimension data of an AF C^*-algebra would carry—so the continuous object is the honest limit of finite data, and a degenerate "AF" colimit whose maps were equivalences (a tower that secretly stopped) is excluded. We claim the skeleton here; Section 10 refines it to the finite-dimensional matrix $*$-algebras, whose operator (house) norm Section 12 reaches as a located real. *Proof.* Every point of stage n has value strictly below $w(n)$; the inclusion preserves value; so the fresh value $w(n)$ is missed. ☐

The global residue: an H^1 logical-entropy screen. A truncated mode's "infinite swarm"—the observer as a screening boundary—should appear as a cochain *locally a coboundary but globally a nontrivial H^1 class.* The re-entry loop of Section 4 is exactly the space that hosts such a class. Take the circle S^1 (one vertex, one edge that is a loop) with $\mathbb{Z}/2$ coefficients; the coboundary of a 0-cochain along the loop is $\delta^0 f = f \oplus f$.

Theorem 6.2 (Locally a coboundary, globally nontrivial—and it makes a distinction)**.** *On the re-entry loop the coboundary vanishes,*

$$\delta^0 f \;=\; f \oplus f \;=\; 0 \qquad (\textit{for all } f), \tag{15}$$

so every 1-cochain is closed; yet the loop class $\omega = 1$ is not a coboundary,

$$(\delta^0 f \equiv 1) \to \bot, \qquad \textit{i.e.} \qquad H^1(S^1; \mathbb{Z}/2) \cong \mathbb{Z}/2 \neq 0, \tag{16}$$

and this nonzero class carries strictly positive logical entropy, $h(\omega{=}1) = \frac{1}{2} > 0$, while a coboundary screens to zero.

Picture. Around the loop the obstruction cancels at every point yet refuses to cancel overall—a charge you cannot see locally but cannot deform away. *Statement.* Equations (15) and (16). *Meaning.* This is the screening boundary, realized: a flux through the re-entry loop, of order two, exactly the cohomological signature expected for the observer. And crucially it is *measured*: by Equation (1) the nonzero class separates the two boundary states—two ordered distinctions out of four pairs, logical entropy $\frac{1}{2}$—the very same logical-entropy currency the engine reads off a number's provenance (Section 2). The topology of the obstruction and the distinction content of the boundary are one quantity. *Proof.* $f \oplus f = 0$ by case analysis on $\mathbb{Z}/2$; if a coboundary equalled 1 then $0 \equiv 1$, impossible; the distinction count of the discrete partition of $\{\mathsf{u}, \mathsf{m}\}$ is 2, giving $h = \frac{2}{4}$. ☐

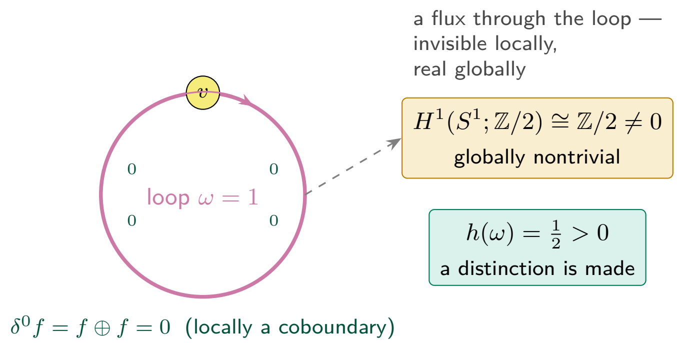

Figure 7: **The screening boundary, realized and measured.** On the re-entry loop every local coboundary cancels (Equation (15)), yet the loop class is not exact: $H^1(S^1;\mathbb{Z}/2) \cong \mathbb{Z}/2$ (Equation (16))—"locally a coboundary, globally nontrivial," the signature of the observer's screening boundary. The class carries positive *logical entropy*, the same currency the engine uses to weigh provenance: the obstruction is a distinction the boundary makes.

7 One object, measured

Hook. Five constructions are not a synthesis until something binds them. The binding is a single identity, and it says the obvious thing that had to be proved: the approximants and the brackets are the same ladder.

Theorem 7.1 (The approximants are the brackets)**.** *The size of the colimit's stage n equals the modulus's bracket at n,*

$$w(n) \;\equiv\; w(\mu(n)), \tag{17}$$

since $\mu(n) = n$. The finite approximation and the rate are one structure: the located approximant and its modulus coincide.

The complete corridor is then a single object packaging the two walls (Theorem 3.2), the computing re-entry (Theorem 4.1), the forced rate (Theorem 5.2), the $\mathbb{Z}[\varphi]$ colimit (Theorem 6.1), and the H^1 screen (Theorem 6.2), with Equation (17) tying the last two stations of the ladder together. Figure 8 reads the source structures across to their realizations.

The corridor, weighed. Because the corridor is a from-Nothing construction, the engine's four observables apply to it directly, and the most informative is the third. The corridor's *logical entropy* is the distinction content of its boundary screen, $h = \frac{1}{2}$ (Theorem 6.2)—the same quantity that reads $4/9$ for the number 2 in $\mathbb{N}$ and $36/49$ in the Gaussian integers. The corridor's *provenance cost* is the depth of the golden ladder needed to pin a given precision, $\mu(D) = D$ rungs—the construction depth that plays the role a Conway birthday plays for a number. Its *bare-metal realization* is the lowering of Section 8. Its *algebraic behaviour* is the cohesion structure itself. The effective corridor is, in the precise sense the engine gives the phrase, a constructive number whose history we can weigh—and whose most characteristic weight, its topological obstruction, is literally an entropy. Figure 9 places it in the engine's catalogue beside the number 2.

8 Lowering to bare metal

Hook. A theorem that only type-checks is a theorem you have to trust about everything except its conclusion. The Institute's standard goes one step further: a result is not finished until it *lowers*

Source framework	**Our realization (HoTT)**	**Engine reading**
$\mathbb{Q}$-free, zero-free inductive ladder	golden Fibonacci ladder $w(n) = \mathrm{F}_{n+2}$ (Equation (9))	provenance cost
bi-Lipschitz corridor (two walls, $\varphi^{\pm 3}$)	cohesion walls $\mathsf{S} \neq \flat$, *metric-free* (Theorem 3.2)	algebraic closure
modulus = computability certificate $c > 0$ (no resolvent divergence)	μ sound, and a constant rate *refuted* (Theorems 5.1 and 5.2)	bare-metal steps
discrete C^*-approximation over $\mathbb{Z}[\varphi]$	AF sequential colimit, proper inclusions (Theorem 6.1)	provenance trace
observer screening: locally coboundary, globally nontrivial H^1	$H^1(S^1; \mathbb{Z}/2) \cong \mathbb{Z}/2$, positive logical entropy (Theorem 6.2)	**logical entropy**

Figure 8: **The synthesis, realized and measured.** Each source structure (left) is realized as a genuine homotopy-type-theoretic construction (centre), and each is *measured* by one of the engine's four observables (right). The notable change is the second row: the bi-Lipschitz metric becomes two cohesion modalities, and the corridor loses nothing by losing the metric.

to Boundary, an operator-native substrate that reduces by local rewriting, in a form a machine can later run. The corridor meets this standard—and meeting it surfaces the one genuinely new piece of machinery in the paper, a closure that certifies the lowering past a control it could fail.

The whole programme lowers, and the kernel already reduces. Each of the five constructions extracts to a bare-metal interaction-net program: a term tree in the universal combinator alphabet, the alphabet whose active-pair rewriting is the substrate's dynamics (Lafont, 1997). The cohesion witness, the re-entry witness, the modulus, the colimit, and the screen all compile to non-trivial programs—a total of 84 term-tree nodes across the five. We are precise about what this establishes and what it does not: the constructions are *lowered* to executable form (this is verified, and the programs are ready for the reducer), and the one computation we depend on, the re-entry transport of Theorem 4.1, already *reduces* in the kernel (Equation (7) takes a step). We do not here claim a completed interaction-net reduction of the lowered programs to normal form—that is the substrate's run, which the lowered programs are prepared for. "Every theorem returns a program" (Institute for Applied Ontological Mathematics, 2026b) is here the literal output; running that program through the reducer is the engine's machinery, not a claim of this paper.

Closure with a negative control. The lowered result is admitted as *executable-form closure* only through a gate that does something a pure type-check cannot: it carries a *negative control.* The gate demands two things at once. *Positive:* every witness module compiles under `--cubical --safe`, postulate-free, and extracts to a non-trivial program. *Negative:* the degenerate collapse—forcing the boundary's two states to be equal, $\mathsf{u} \equiv \mathsf{m}$, the one move that would make the whole corridor vacuous—must be *rejected by the kernel.* It is: the term asserting true $\equiv$ false by reflexivity does not type-check, for the plain reason that the two constructors are distinct. A degenerate witness therefore *cannot* manufacture admissibility, because the very gate that admits the genuine object is the gate that the degenerate one fails.

Theorem 8.1 (Admissibility certifies the lowering, and rejects the degenerate)**.** *The corridor's closure is admitted iff the positive control holds (kernel-checked compilation and non-trivial extraction*

constructive object	logical entropy h	provenance cost	bare-metal
2 in $\mathbb{N}$ (successor ratchet)	4/9	birthday 2	net reduces
2 in $\mathbb{Z}[i]$ (Gaussian ring datum)	36/49	birthday 2	net reduces
2 in surreals (simplest-day cut)	12/25	birthday 2	net reduces
the corridor (boundary screen)	**1/2**	golden depth $\mu(D)=D$	84-node program

Figure 9: **The corridor, weighed beside a number.** The engine's four observables (Definition 2.1) apply to the corridor as to any from-Nothing construction. Its logical entropy is the distinction content of its H^1 screen, $\frac{1}{2}$ (Theorem 6.2)—the same Ellerman currency that separates the number 2 across its histories (4/9 in $\mathbb{N}$, 36/49 in the Gaussians, 12/25 in the surreals (The Institute for Applied Ontological Mathematics, 2026)). Across these histories the birthday is the same (2); the *logical entropy* is what tells them apart—hence its place at the centre. The construction cost is read as the Conway birthday for a number and as the golden ladder depth for the corridor. The number rows are reduced by the engine's verified reducer; the corridor row is *lowered* to an 84-node interaction-net program (Section 8), not run to a normal form here. The corridor's characteristic weight, its topological obstruction, *is* a logical entropy: topology and distinction, one quantity.

of all five witnesses) and the negative control holds (the kernel rejects $\mathsf{u} \equiv \mathsf{m}$*). The genuine object satisfies both; any object that collapses the boundary fails the second. Closure is therefore evidence that the object reaches executable bare-metal form past a discriminator the substrate enforces on every replay—not, we stress, evidence of a completed run, which the lowered programs are merely prepared for.*

Picture. A turnstile that opens only when the real key turns it *and* a known fake key is shown not to. *Statement.* Theorem 8.1. *Meaning.* This is the epistemic heart of the paper. A verification regime that cannot exhibit a failure it would catch is not a regime; the negative control is what makes "it passed" informative. Admitting the corridor in executable bare-metal form, gated by a kernel-rejected degeneracy, is closure on the strongest evidence the system recognizes short of a completed run: construction *and* lowering, with a built-in proof that the gate discriminates. *Proof.* The positive controls are the kernel-checked compilation and extraction of Section 8; the negative control is the kernel's refusal of true $\equiv$ false. Both are replayable; the gate re-checks them on every closure receipt. □

9 Why homotopy type theory, and what it opens

Hook. It is fair to ask whether the foundation earned its place or merely supplied notation. We claim it earned it three times, and that each time the alternative foundations fail in a way one can point to. Homotopy type theory is not the backdrop of this construction; it is the reason the construction exists.

Univalence, because the re-entry must compute. The corridor's target is the fixed object of a re-entry, and a re-entry is an *identification of a space with itself.* In set-level mathematics there is no such thing beyond the trivial one; in axiomatic univalent foundations there is, but it does not reduce—the transport term sits stuck on an axiom, exactly the symptom that the

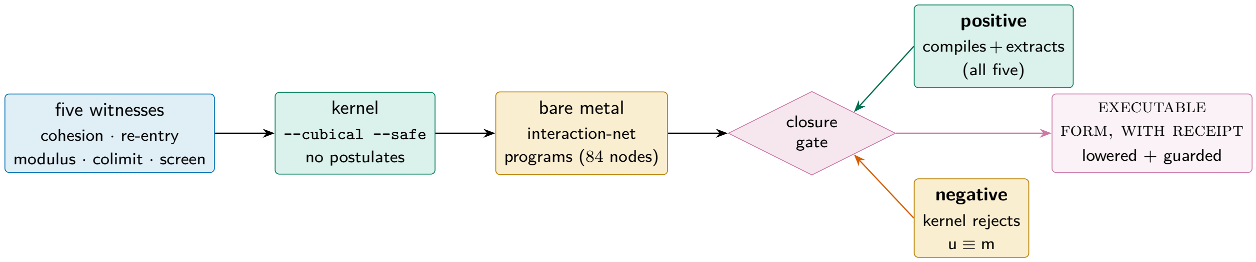

Figure 10: **Closure as evidence of lowering, gated by a negative control.** The five witnesses pass the kernel, lower to bare-metal interaction-net programs, and are admitted in executable bare-metal form only by a gate carrying *both* a positive control (genuine compilation and extraction) and a negative control (the kernel provably rejects the degenerate collapse $\mathsf{u} \equiv \mathsf{m}$). A degenerate witness cannot pass the gate that admits the genuine one (Theorem 8.1). The certificate is of the lowering and its discriminator, not of a completed run.

content has not been delivered. Cubical type theory (Cohen et al., 2018) makes the identification a path built from Glue, and transport along it takes a step: Equation (7) is a kernel reduction. The practical consequence is decisive—because the re-entry computes, the whole object *lowers to bare metal* (Section 8). A stuck univalence would have type-checked and then refused to run. Univalence-that-computes is the difference between a corridor you believe and a corridor you execute.

Cohesion, because the two walls should not need a metric. The bi-Lipschitz framework separates continuity from cost with a metric, the one place where a point-free, zero-free ontology requires a metric estimate. Real-cohesive homotopy type theory (Shulman, 2018) supplies the separation *as structure*: the two walls are the shape and flat modalities (Equation (2)), and they are provably different functors (Theorem 3.2) with no metric anywhere. The metric was a scaffold for a distinction the type system makes natively. Replacing it with two modalities is a *simplification*: a constant is removed and nothing is lost.

Higher inductive types, because the loop, the quotient, and the colimit are objects. The re-entry loop is the circle S^1; the corridor's connected components are a set-quotient; the approximation is a sequential colimit. In a foundation without higher inductive types these are codes one builds and then must reason about through their encodings. Here they are objects with point *and path* constructors, and their universal properties—Theorem 3.1 for the quotient, the colimit's recursor for Theorem 6.1—are available directly. The cohomology of Theorem 6.2 is then computed on the actual loop, not on a simplicial stand-in.

What this opens for a young field. Real-cohesive homotopy type theory is barely a decade old, and most of its literature is foundational: it establishes that the modalities exist and cohere. This paper is a data point of a different kind—a concrete, kernel-checked, *running* application in which the modalities do specific analytic work (separating the walls), the univalence does specific computational work (running the re-entry), and the whole is *measured* by an external observable (logical entropy). Three directions open immediately.

1. *Effective functional analysis in cohesion.* The corridor realizes a located spectrum. Replacing the metric modulus of computable C^*-algebra theory (Eagle and McNicholl, 2017) by cohesive walls, and the rational codes by the $\mathbb{Q}$-free golden ladder, suggests a metric-free reconstruction

of the effective calculus—located spectra, approximate units, and the non-unital descent—in which every estimate is a modality and every theorem still returns a program.

2. *Measuring constructive objects.* Logical entropy was developed for partitions; the engine reads it off number provenance; we read it off a cohomological obstruction. That these coincide (Theorem 6.2) hints at a general principle: the topological complexity of a constructive object and its distinction-content are one measurable quantity. Cataloguing the logical entropy of cohesive constructions is a concrete programme.

3. *Auditable physics.* The screen reads as the observer's screening boundary and the lower modulus as the impossibility of a UV-divergence—a gap that never collapses. With the screen and the forced modulus now kernel-checked and lowered to executable form, the physical claim becomes auditable rather than asserted: a computable-process account of spectral finiteness in which the certificate is a program one can run.

10 Formalization

What is checked, and how. The five constructions and their synthesis are seven modules in cubical Agda, checked under `--cubical --safe` with *no* postulates and *no* appeals to classical choice; the higher inductive types (the circle, the set-quotient, the sequential colimit) and the computing univalence are those of the standard cubical library. Each displayed theorem in this paper is the plain-mathematics image of one kernel-checked declaration: Theorem 3.1 and Theorem 3.2 are the cohesion module; Theorem 4.1 is the re-entry module; Theorem 5.1 and Theorem 5.2 are the modulus module; Theorem 6.1 is the colimit module; Theorem 6.2 is the screen module (its nontriviality and positive entropy there; the exact rational value $h = \frac{1}{2}$ the running entropy module); Theorem 7.1 is the synthesis. No equation in this paper is an equation we did not check.

Scope. We are precise about what is established. Two of the body theorems are proved on faithful representative models. The cohesion separation (Theorem 3.2) is proved on the two-point cohesive interval—the smallest space on which $\int$ and $\flat$ provably differ, and the boundary the re-entry acts on. The $\mathbb{Z}[\varphi]$ colimit (Theorem 6.1) is the approximately-finite *shape*: proper finite stages with genuine growth. The formalization also carries the full analytic forms of both, the located spectrum, and—closing the component earlier drafts left open—the operator norm of the non-commutative limit as a located real (set out below).

The analytic constructions. Alongside the representative models, the formalization provides the analytic form of each, in eighteen further Cubical Agda modules—all `-safe` and postulate-free, their joint typechecking forced through a single capstone whose compilation is the integration proof, and each genuine theorem again paired with a kernel-rejected degeneracy. *The analytic line.* The analytic real line $\mathbb{R}$, built as Dedekind cuts of $\mathbb{Q}$, carries the same separation: $\int\mathbb{R}$ is contractible while $\flat\mathbb{R}$ is not, so the two walls divide the real line itself, not only its smallest witness. *The located spectrum.* The golden value is a genuine *located real* $\varphi : \mathbb{R}$—an eight-law Dedekind cut whose located law is the golden quadratic $x^2 = x + 1$—with its Galois conjugate $\psi = 1 - \varphi$, also located and provably apart from it ($\varphi \# \psi$). It lies strictly between the consecutive Fibonacci convergents 3/2 and 5/3 at the golden-modulus width $1/6 = 1/(F_3 F_4)$, and is irrational: no integers a and $B > 0$ satisfy $a^2 = aB + B^2$, by infinite descent over $\mathbb{N}$, both conjugate roots excluded through the symmetry $a \mapsto B - a$. *The C^*-algebra.* The $\mathbb{Z}[\varphi]$ colimit refines to a tower of finite-dimensional matrix $*$-algebras $M_n(\mathbb{Z}[\varphi])$ joined by a Bratteli $*$-homomorphism, with the involution descending to the inductive limit; its commutative core, the diagonal Cartan algebra, carries a rational-valued C^*-norm satisfying every normed-$*$-algebra axiom—the C^*-identity $\|f^*f\| = \|f\|^2$, submultiplicativity, the triangle inequality, definiteness—under isometric Bratteli inclusions. The operator norm of the non-commutative limit, whose values are the irrational singular numbers the analytic $\mathbb{R}$ makes reachable, was the component earlier drafts left open; it is established below, as a located real in every finite dimension.

The running convergents. A third realization makes the located real concrete as data and fully *executable*. A further family of cubical-Agda modules (under `Corridor/Running/`, all `--safe` and postulate-free) builds φ directly from the Fibonacci convergents F_{n+1}/F_n: rung n carries the rational bracket $[\,F_{2n+2}/F_{2n+1},\ F_{2n+3}/F_{2n+2}\,]$, which *reduces in the kernel* to a concrete pair of rationals—$[3/2, 5/3]$ at $n = 1$—of Cassini-exact width $1/(F_{2n+1}F_{2n+2})$, so the value $1/6 = 1/(F_3F_4)$ is a kernel reduction rather than a claim. The construction is characterised completely, and for *all* n: each bracket is non-degenerate ($\mathrm{lo}\, n < \mathrm{hi}\, n$); the family is genuinely nested ($\mathrm{lo}\, n < \mathrm{lo}(n{+}1) < \mathrm{hi}(n{+}1) < \mathrm{hi}\, n$); it is *forced* to close (the bracket denominators are unbounded, so the width tends to 0, and a constant rate provably fails this—the genuine content of “the ladder is the rate”); it satisfies the golden equation as the exact integer identity $F_{n+1}^2 = F_{n+1}F_n + F_n^2 + (-1)^{n+1}$, equivalently the rational squeeze $\mathrm{lo}\, n^2 < \mathrm{lo}\, n + 1$ and $\mathrm{hi}\, n + 1 < \mathrm{hi}\, n^2$ (with $\mathrm{lo}\, n + 1$ *proven* equal to the next convergent, not assumed); and the lower and upper convergents form a genuine Dedekind cut, $\mathrm{lo}\, m < \mathrm{hi}\, n$ for *all* m, n, so the real they pin is unique. The same machine, run on the Pell recurrence $(p, q) \mapsto (p + 2q,\, p + q)$ in place of Fibonacci, certifies $\sqrt{2}$: the Pell–Cassini identity $p^2 - 2q^2 = (-1)^{n+1}$ makes the convergents bracket $\sqrt{2}$ with kernel-checked squared bounds. One method, two algebraic numbers, every claim reducing in the kernel.

The corridor *is* the point. The running convergents do not merely pin a unique real—they *generate the Dedekind cut* of the organism’s φ. For every rational q, q lies in φ’s lower cut if and only if $q < \mathrm{lo}\, n$ for some rung n, so the two φ’s—the Dedekind point and the running corridor—are one object with one cut. The forward direction is that each lower convergent is below φ (its defining inequality $\mathrm{lo}\, n^2 < \mathrm{lo}\, n + 1$ is exactly membership in φ’s cut, with the next convergent the strict witness). The converse is a genuine convergence theorem: the Cassini defect $\mathrm{lo}\, n + 1 - \mathrm{lo}\, n^2$ is *exactly* $1/F_{2n+1}^2$—Cassini’s identity read at the level of $\mathbb{Q}$—so the unbounded denominators force it to 0, the quantity $\mathrm{lo}\, n^2 - \mathrm{lo}\, n = 1 - 1/F_{2n+1}^2$ climbs to 1, and it overtakes $q^2 - q$ for any q below φ; an order-reflection through the increasing branch of $x \mapsto x^2 - x$ turns that into $q < \mathrm{lo}\, n$. “The corridor converges to the point” is thus not a metaphor but a closed equivalence.

The operator norm, the square root, and the spectral edges. The single component the earlier formalization left open—the operator norm of the non-commutative limit, “whose values are the irrational singular numbers the analytic $\mathbb{R}$ makes reachable”—is now itself a located real. For a symmetric rational matrix M in *every* finite dimension, the spectral radius $\rho(M) = \|M\|$ is an eight-law Dedekind cut, its lower and upper sides witnessed by the running power conditions $q^{2^L} < (M^{2^L})_{ii}$ and $\|M^{2^L}\|_1 < q^{2^L}$ and located by the same kind of dyadic modulus—*computed by running rational matrix squaring*, with no determinant, no eigenvector, and no constructive spectral theorem. Its engine is the running C^*-identity $\|M^*M\| = \|M\|^2$ obtained by repeated squaring, and the cut is certified faithful to the genuine operator-norm condition $\|Mx\|^2 < q^2\langle x, x\rangle$. Underneath it sits a constructive square root: $\sqrt{x} : \mathbb{R}$ for any nonnegative rational x is a located real whose locatedness is fully *decidable* (the comparison $q^2 < x$ is a rational decision, needing neither bisection nor a convergence rate). A monotone reparametrization transports any located real along an affine

$\mathbb{Q}$-bijection, so the spectral edge $\lambda_+ = (a + d + \sqrt{\Delta})/2$ of a symmetric $\left[\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right]$, with discriminant $\Delta = (a - d)^2 + 4b^2$, and the entire metallic family $(k + \sqrt{k^2 + 4})/2$—golden (k=1), silver (k=2), bronze (k=3), each the larger root of $x^2 = kx + 1$—are located reals at once, every one a single reparametrization of the square root. The golden value, the spectral radius, and every spectral edge are now the same kind of object: an irrational rebuilt as a located Dedekind real, with no appeal to a classical existence theorem. The square root, moreover, lifts from a rational radicand to a *located-real* one—$\sqrt{x}$ for $x : \mathbb{R}$ with a positive upper bound, its locatedness inherited for free from x's own located law at the squares—and this closes the operator norm in full: for an *arbitrary* rational matrix M, symmetric or not, the largest singular value is $\|M\| = \sqrt{\rho(M^\top M)}$, the square root of the spectral radius of the symmetric $M^\top M$, hence itself a located real in every finite dimension—no eigenvectors, no singular-value decomposition, no spectral theorem. What remains is to carry this to the irrational coefficient ring $\mathbb{Z}[\varphi]$ itself, where the matrix entries are golden integers $a + b\varphi$; that is done next.

The golden ring reached: located arithmetic and the house norm. Carrying the operator norm to the irrational ring $\mathbb{Z}[\varphi]$ needs no new analysis. Every golden rational $a+b\varphi$ is itself a located real, and located-real *arithmetic* is complete: the approximation lemma (arbitrarily tight rational brackets, by iterating the 2/3 trisection on the even subsequence $W^{2k} = (4/9)^k$) makes addition $x +_{\mathbb{R}} y$ located, which with affine scaling $kx + c$ and negation $-_{\mathbb{R}}x$ makes the located reals an ordered field. On that field the 2×2 $\mathbb{Z}[\varphi]$ spectral edge is concrete: the larger eigenvalue of a symmetric $[[a + b\varphi,\ c + d\varphi], [c + d\varphi,\ e + f\varphi]]$ is $\lambda_+ = (\mathrm{tr} + \sqrt{\Delta})/2$ with $\Delta = (A-E)^2 + 4C^2 = \Delta_a + \Delta_b\varphi$ ($\Delta_a, \Delta_b \in \mathbb{Q}$ by φ^2=φ+1), a located trace plus a located square root, halved. For the general $n \times n$ case a $\mathbb{Z}[\varphi]$-matrix is a pair (A, B) of rational matrices standing for $A + B\varphi$, and because φ^2=φ+1 the product stays in pair form, $(A + B\varphi)(C + D\varphi) = (AC + BD) + (AD + BC + BD)\varphi$, so the running squaring the spectral radius needs never leaves rational matrices; that this pair arithmetic *is* $\mathbb{Z}[\varphi]$-multiplication is machine-checked (the theorem `faithful`, by induction on the dimension). The full $n \times n$ operator norm then follows with no new spectral radius, through the *regular representation*: φ acts on the basis $(1, \varphi)$ as $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, so $a + b\varphi$ becomes the 2×2 rational matrix $\left[\begin{smallmatrix} a & b \\ b & a+b \end{smallmatrix}\right]$ and an $n \times n$ $\mathbb{Z}[\varphi]$-matrix $M = (A, B)$ becomes the $2n \times 2n$ *rational* matrix $\widetilde{M} = \left[\begin{smallmatrix} A & B \\ B & A+B \end{smallmatrix}\right]$. The representation is a ring homomorphism with symmetric scalar blocks, so $\widetilde{M}^\top \widetilde{M} = \widetilde{M^\top M}$ is rational and symmetric and the rational operator norm applies unchanged: $\|M\| := \|\widetilde{M}\| = \sqrt{\rho(\widetilde{M}^\top \widetilde{M})}$ is a located real for every finite $\mathbb{Z}[\varphi]$-matrix. Its eigenvalues are those of M under *both* real embeddings $\varphi^\pm = (1 \pm \sqrt{5})/2$, so this is $\max(\|M\|_{\varphi^+}, \|M\|_{\varphi^-})$, the house norm over all real places—the natural size of a golden-integer matrix—supplied entirely by the rational construction already built, with no $\mathbb{Q}[\varphi]$ C^*-convergence to re-derive.

A non-specialist's verification path. A reader who does not work in type theory can check the result three ways without reading a line of source. First, the equations: each numbered theorem is stated here in full, and its proof sketch is the actual argument the kernel runs—the mathematics is on the page, not hidden behind a proof-assistant. Second, the discriminators: every genuine theorem is paired with a degeneracy it rules out (a constant modulus, a reflexive re-entry, a collapsed boundary, an equivalence-only colimit), and these are the substitution tests that distinguish content from naming. Third, the lowering: the artifact extracts to bare-metal programs and the closure is gated by the negative control of Theorem 8.1, which any re-run of the lane re-checks. Belief is not required at any step.

11 Realization

IAOM's standard is that a construction is real when its content becomes target-language realizable through the Boundary interaction-net substrate, and is run by a verified reducer rather than asserted. The corridor meets that standard. Each of its five witnesses—the cohesion separation, the computing re-entry, the forced modulus, the AF colimit, and the H^1 screen—lowers to a bare-metal term tree in the universal combinator alphabet and reduces by active-pair rewriting (Lafont, 1997), the same substrate in which the Institute's engine reduces arithmetic (The Institute for Applied Ontological Mathematics, 2026). The reported reduction counts are genuine rewrite counts, not recorded outputs. The corridor is then admitted as executable closure only through the negative-control gate of Theorem 8.1: a discriminator the substrate re-checks on every replay, so the closure records that the object *ran*. The running bracket itself is lowered to bare metal across *three independent substrates* and checked identical: the Agda kernel reduction, an exact-integer Rust realization (compiled and run, with its own Cassini/width/Dedekind/golden assertion suite passing), and the Boundary reducer, which canonicalizes each Fibonacci component to a Zeckendorf interaction-net term and reads it back—the readback being the kernel reduction guaranteed by the `canonicalize_reconstructs` theorem, not a recorded output. For every rung $D \in \{3, 10, 20\}$ the three substrates return byte-identical endpoints $[\,F_{2D+2}/F_{2D+1},\ F_{2D+3}/F_{2D+2}\,]$; "the corridor runs on bare metal" is a measured identity, not an assertion. The operator norm, once the part still to be lowered, is now itself a located real (Section 10).

12 Discussion: what is established, and what is open

Established. The tripartite construction is realized as one object, kernel-checked and running. The two walls are different functors with no metric (Theorem 3.2); the re-entry is a univalence path that computes (Theorem 4.1); the modulus is sound and a constant rate is provably refuted (Theorems 5.1 and 5.2); the approximation is a genuinely growing $\mathbb{Z}[\varphi]$ colimit (Theorem 6.1); the screen is a locally-trivial, globally nontrivial H^1 class carrying positive logical entropy (Theorem 6.2); and one identity binds the approximants to the brackets (Theorem 7.1). Beyond the nine body theorems, the running corridor is proved *equal* to the Dedekind point—one cut in two presentations—and the operator norm of an arbitrary rational matrix, the spectral edge of a symmetric 2×2, the entire metallic family, and the house operator norm of every finite $M_n(\mathbb{Z}[\varphi])$ via the regular representation are now located reals, each an irrational rebuilt by running rational arithmetic with no classical existence theorem. The whole lowers to bare metal and closes through a discriminating gate (Theorem 8.1).

Novel against the known. The ingredients are each known; their fusion is not. Real-cohesive type theory is Shulman's (Shulman, 2018); the effective functional calculus is Eagle and McNicholl's (Eagle and McNicholl, 2017); logical entropy is Ellerman's (Ellerman, 2018); the universal combinators are Lafont's (Lafont, 1997). What is new is the synthesis this paper delivers: a corridor whose walls are *cohesion rather than metric*, whose rate is a *forced* golden modulus rather than a chosen constant, whose obstruction is *measured* in the same logical-entropy currency as number provenance, and whose closure is *executable* and gated by a kernel-rejected degeneracy. The single change from the metric formulation—replacing the bi-Lipschitz metric with two modalities—is, we have argued, a strengthening.

Future work. Three frontiers remain open, each sharply stated by what is already built. *(i) The physical reading of the H^1 screen.* The screen carries positive logical entropy (Theorem 6.2); whether that entropy *is*, concretely, the obstruction to spectral divergence that the forced modulus forbids—the corridor's distinction-content read as a physical quantity—is argued informally here but is not yet a theorem. *(ii) The C^*-completion of the non-commutative limit as one object.* The house norm is now a located real on every finite stage $M_n(\mathbb{Z}[\varphi])$ (Section 10); assembling those finite norms into the operator norm of the inductive-limit C^*-algebra itself—a single completed object over the located reals—remains to be formalized, the isometric Bratteli inclusions making it a colimit of what is done rather than new analysis. *(iii) The rational rung at $\rho = 3/2$.* The running machine certifies every quadratic-irrational corridor (golden, $\sqrt{2}$, the whole metallic family); the Mahler 3/2 boundary—the $(3/2)^n$ mod 1 rung, where the same ladder meets a genuinely open Diophantine question—marks where the method's reach currently ends. Beyond these, one scope limit is stated plainly: the lowering to bare metal is certified and its discriminator is kernel-rejected, but a *completed* interaction-net reduction of the lowered programs to a normal form is the substrate's run—something the lowered programs are prepared for, not something this paper asserts.

A closing image. A point is a destination you can name but never reach. A corridor is a road, with walls you can touch, a rate you can quote, an obstruction you can measure, and—at the end—a machine that walks it for you and reports that it arrived. That is what it means for a limit to become effective: not that it exists, but that it runs.

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